DIRECTIONS *for questions 1 to 5:* Answer these questions on the basis of the information given below.

During a particular month, each of seven persons, A through G, met one or more of the other six persons. For any pair of persons, say X and Y, if X met Y, then Y is also considered to have met X.

It is also known that

i. A met all the persons that F met, not considering A and F.

ii. A and B did not meet each other and among the persons that B met, A did not meet exactly two persons.

iii. C met E and two other persons, but did not meet B, while D met more number of persons than C.

iv. the number of persons that the seven persons met are 1, 2, 2, 3, 3, 4 and 5, not necessarily in any specific order.

v. G did not meet any person who met F, while E met both B and F.

Q1. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
How many persons did G meet?

a) **1**

b) **2**

c) **Either 2 or 3**

d) **Either 1 or 3**

Q2. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
Who among the following met the highest number of persons?

- a) **E**
- b) **D**
- c) **B**
- d) **G**

Q3. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
Who among the following did A meet?

- a) **D**
- b) **C**
- c) **F**
- d) **Both F and C**

Q4. **DIRECTIONS** *for questions 1 to 5:* Select the correct alternative from the given choices.
Which of the following persons did B meet but C did not meet?

- a) **A**
- b) **D**
- c) **G**
- d) **None of the above**

Q5. DIRECTIONS *for questions 1 to 5:* Select the correct alternative from the given choices.
How many persons did both E and D meet?

- a) **4**
- b) **1**
- c) **3**
- d) **2**

DIRECTIONS *for questions 6 to 10:* Answer these questions on the basis of the information given below.
Kiran was analysing the revenues of thirty companies in an industry. He ranked the companies from 1 to 30, in the descending order of their revenues. The revenues (in $ million) of the thirty companies were distinct positive integers.

The following table provides the average revenues of the companies that have ranks in different ranges:

Rank of Companies	Average Revenue (in $ million)
1 to 5	18145.2
6 to 10	9762.0
11 to 15	5996.8
16 to 20	4137.4
21 to 25	3359.0
26 to 30	2734.0

For example, the average revenue of the five companies that have ranks from 1 to 5 (both inclusive) is $18145.2 million.

It is also known that for any values of n and m, where n > 3 and m is a positive integer, the revenue of the company that was ranked n was m times the revenue of the company that was ranked m × n, provided m × n is not greater than 30

Q6. DIRECTIONS *for questions 6 to 8:* Type in your answer in the input box provided below the question.
What is the revenue (in $ million) of the company ranked 21?

Q7. DIRECTIONS *for questions 6 to 8:* Type in your answer in the input box provided below the question.

What is the difference (in $ million) between the revenue of the company ranked 18 and the revenue of the company ranked 27?

Q8. DIRECTIONS *for questions 6 to 8:* Type in your answer in the input box provided below the question.
What is the average (in $ million) of the revenues of the companies that have a rank greater than 13 and less than 17?

Q9. DIRECTIONS *for questions 9 and 10:* Select the correct alternative from the given choices.
Which of the following BEST represents the revenue (in $ million) of the company ranked 11?

- a) **It lies between 6944 and 7560.**
- b) **It lies between 6300 and 7199.**
- c) **It is exactly 7196.**
- d) **It is exactly 7198.**

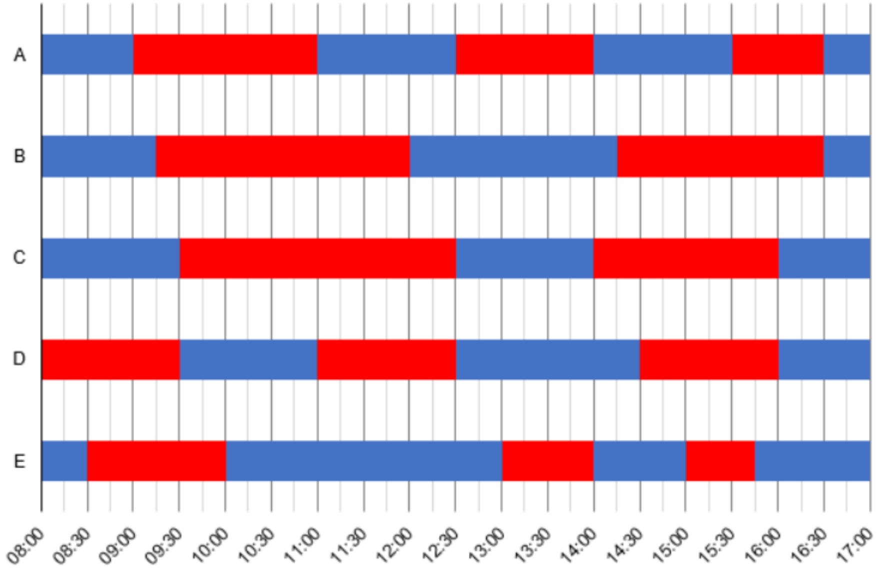
Q10. **DIRECTIONS** *for questions 9 and 10:* Select the correct alternative from the given choices.
What is the difference (in $ million) between the revenue of the company ranked 23 and the revenue of the company ranked 29?

- a) **805**
- b) **795**
- c) **810**
- d) **815**

DIRECTIONS *for questions 11 to 15:* Answer these questions on the basis of the information given below.

Five machines – A through E – manufacture widgets in a factory. Each machine can be run in two modes – Normal and Turbo. When running in Normal mode, any machine can manufacture 2 widgets per minute. However, when running in Turbo mode, any machine can manufacture 6 widgets per minute. Further, for running in Normal mode, any machine consumes a fuel of 30 ml/minute; and for running in Turbo mode, any machine consumes a fuel of 180 ml/minute. The machines are operated every day from 8:00 to 17:00.

The chart below shows, for a particular day, the times at which each of the machines was operated in Normal mode and in Turbo mode. The times at which any machine was operated in Normal mode is shaded in blue, while the times at which any machine was operated in Turbo mode is shaded in red. Assume that any machine can change modes instantly.



Q11. **DIRECTIONS** *for questions 11 to 15:* Select the correct alternative from the given choices.

What is the difference between the total number of widgets manufactured by machine A and that by machine C during the day?

- a) **60**
- b) **120** ✓ Your answer is correct
- c) **360**
- d) **240**

Q12. **DIRECTIONS** *for questions 11 to 15:* Select the correct alternative from the given choices.
Which machine consumed the least fuel till the end of 12:09?

- a) **Machine B**
- b) **Machine C**
- c) **Machine E** ✓Your answer is correct
- d) **Machine A**

Q13. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

What is the difference (in ml) between the fuel consumed by machine C until it completed the manufacturing of its 1000^{th} widget of the day and that by machine E until it completed the manufacturing of its 1000^{th} widget of the day?

- a) **3000**
- b) **3200**
- c) **3400**
- d) **3600** ✓ **Your answer is correct**

Q14. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

If the number of widgets manufactured by each of the five machines in the first n minutes of the day (counting from 8:00 AM) was at least 4n, which of the following BEST represents all the possible values of n?

- a) **180 or 240 or 300**
- b) **Exactly 180**
- c) **Lies between 240 and 300**
- d) **Lies between 120 and 300**

Q15. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

If the number of widgets manufactured by machine B after consuming n ml of fuel is at most n/25, what is the minimum possible value of n?

- a) **11250**
- b) **7500**
- c) **450**
- d) **300**

DIRECTIONS *for questions 16 to 20:* Answer these questions on the basis of the information given below.

Eight persons, A through H, were playing a card game sitting in eight chairs, which were arranged equally spaced around a circular table. The eight persons played exactly seven rounds of the game such that each round was won by exactly one person and in each round exactly one person was knocked out of the game. The person who won the last round of the game is said to be the winner of the game. At the end of any round, the person who is knocked out vacates his chair and the rest remain seated in their chairs. Any round is said to have ended only after the person who was knocked out at the end of that round vacates his chair. Also, at the beginning of the game, the eight persons, A through H, had exactly ₹700, ₹500, ₹600, ₹250, ₹150, ₹400, ₹800 and ₹550, respectively.

At the end of any round, the entire amount with the person who was knocked out in that round was transferred to the person who won the round, while the amounts with all the others remained unchanged.

It is also known that

i. after a certain number of rounds, D had more than ₹500 with him and D was not sitting in the chair adjacent to B during the game.

ii. at the beginning of the fourth round, C was sitting in the chair adjacent to the one in which E was sitting, and C won at least one round but did not have more than ₹1200 at any point in the game.

iii. at the end of the fifth round, A and H were sitting in chairs which were opposite each other and both the chairs adjacent to the one in which A was sitting were empty.

iv. the person who was knocked out at the end of the third round was not B and this person had the same amount of money with him throughout the game, until he was knocked out.

v. the two persons who played in the seventh round were sitting in chairs which were adjacent to each other and the last two rounds were won by different persons.

vi. B was sitting to the left of F at the start of the game, while the person who won the second round was sitting opposite a person who did not win any round.

vii. F, who did not win the game, played in the fifth round, while H won at most one round.

viii. no person won two consecutive rounds.

Q16. DIRECTIONS *for questions 16 and 17:* Select the correct alternative from the given choices.
Who won the second round of the tournament?

- a) **F**
- b) **A**
- c) **C**
- d) **E**

Q17. DIRECTIONS *for questions 16 and 17:* Select the correct alternative from the given choices.
Among the players who played in the fifth round, who had the lowest amount at the beginning of the fifth round?

- a) **A**
- b) **F**
- c) **H**
- d) **G**

Q18. DIRECTIONS *for questions 18 and 19:* Type in your answer in the input box provided below the question.
What is the total amount (in ₹) with A at the end of the third round?

Q19. **DIRECTIONS** *for questions 18 and 19:* Type in your answer in the input box provided below the question.

Before the beginning of the seventh round, what is the highest amount (in ₹) with any person during the game?

Q20. DIRECTIONS *for question 20:* Select the correct alternative from the given choices.
Who was sitting opposite the person who won the fourth round?

a) **E**

b) **F**

c) **B**

d) **D**